

Embedded Electronic System for Diesel Level Monitoring in Emergency Power Plants

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Abstract—This project involves technical knowledge in microcontroller programming, PCB design, and manufacturing, as well as the implementation of Internet of Things (IoT) technologies. The main objective was to design and develop an embedded electronic board capable of monitoring the level of diesel in the tank of an emergency power plant, in order to prevent operational and safety risks caused by low fuel levels, possible leakages, or unexpected failures in the backup power system. For the implementation of the system, an ESP32 microcontroller was used together with an ultrasonic sensor JSN-SR04T, in charge of measuring the distance between the sensor and the surface of the fuel. This measurement allows the system to calculate both the percentage of fuel remaining and the available volume inside the tank. In addition, an OLED display and an RGB LED were integrated to provide local and immediate visualization of the system status and fuel level. As part of the IoT application, the card transmits tank level data via Wi-Fi to the Ubidots platform, where the information can be viewed in real time remotely. This enables continuous supervision of the fuel supply and improves maintenance planning. Overall, the system offers a reliable, flexible, and scalable solution for continuous fuel monitoring in emergency plants, with potential for future upgrades and additional sensing capabilities.

Index Terms—Monitoring system, diesel, ESP32, Ultrasonic sensor, PCB design.

I. INTRODUCTION

Emergency power plants constitute a fundamental backup system for industrial, academic, and service electrical facilities, as they guarantee the continuity of the power supply in the event of failures in the electrical grid. These plants operate using fuel, making it essential to continuously monitor the available supply level. In this regard, fuel measurement is important, since leaks or losses may occur, increasing safety risks and affecting the operation of the system.

Therefore, the development of a monitoring system for the diesel tank of the emergency power plant is proposed. This system will employ an ESP32 microcontroller and a JSN-SR04T ultrasonic sensor, which allows the determination of both the tank volume and the filling percentage. Likewise, the system integrates an OLED display and an RGB LED, through which the current diesel level percentage is physically displayed.

As part of the implementation, the Wi-Fi module incorporated into the ESP32 is used to send the information to a cloud platform, where the data can be visualized in real time [7]. This enables the integration of the system within the Internet

of Things (IoT) approach, facilitating remote monitoring and continuous supervision of the tank.

A. Problem statement

Emergency power plants are essential systems to ensure the continuity of the electrical supply in industrial, academic, and service facilities in the event of possible failures in the power grid. Since their operation depends on the use of diesel fuel, it is essential to maintain precise control of the level stored in the tank, as the operational availability of the system in critical situations depends on it.

Currently, fuel level supervision is usually carried out manually or through periodic inspections, which may lead to delays in the detection of low levels and difficulties in identifying leaks or fuel losses. This situation represents a significant risk, since an undetected decrease in diesel fuel may compromise the operation of the emergency power plant precisely at the moment when its use is required.

Likewise, the lack of a real-time monitoring system limits the capacity for remote supervision and the timely decision-making related to preventive maintenance and fuel supply management. This increases safety risks, raises operational costs, and may affect the reliability of the electrical backup system.

For this reason, there is a need to develop an intelligent monitoring system capable of providing real-time information about the diesel level in the tank, facilitating anomaly detection and improving the safety, reliability, and operational efficiency of the emergency power plant.

II. OBJECTIVES

Develop an embedded electronic board capable of measuring diesel levels remotely in emergency plants.

- Investigate and study all the components needed for this project.
- Establish the communication between the ESP32 and the ultrasonic sensor.
- Validate the functionality of the prototype.
- Connect the ESP32 to internet via WiFi.
- Visualize and monitor data through the Ubidots platform.
- Perform functional tests and validation on the developed PCB.

These objectives were part of the project I developed. My partner was in charge of designing the schematic, PCB, and the enclosure for this embedded electronic board.

A. Ultrasonic sensor JSN-SR04T

The JSN-SR04T ultrasonic sensor is a waterproof distance measurement module, ideal for outdoor applications, with a detection range from 25 cm to 450 cm (up to 6 m depending on the version) and an accuracy of ± 3 mm [2]. It operates at 5 V DC, uses a 40 kHz waterproof probe, consumes 30 mA, and is compatible with Arduino through Trig and Echo pins. Its performance is not affected by sunlight; however, soft materials such as fabric may not reflect the ultrasonic waves effectively.



Fig. 1. Ultrasonic sensor JSN-SR04T

B. ESP32

This is a widely used microcontroller due to its ease of programming and low cost. It is a microcontroller designed by the brand Espressif and we chose the ESP32 because it already includes an integrated Wi-Fi module, which is very useful for the IoT implementation that will be carried out with Ubidots [1]. Many low-cost microcontrollers do not include this feature (even the Arduino Uno does not have it), which is why this microcontroller is the most suitable choice for this project.

C. OLED Display

The 128×64 pixel OLED display with an I2C interface is a widely used display module in projects involving Arduino and other microcontrollers due to its low power consumption and ease of connection [4]. It generally incorporates the SSD1306 controller, which allows the display of text, numbers, symbols, and simple graphics with excellent sharpness and contrast [3]. Its resolution of 128 columns by 64 rows of pixels provides a clear visualization of data, menus, and system variables.

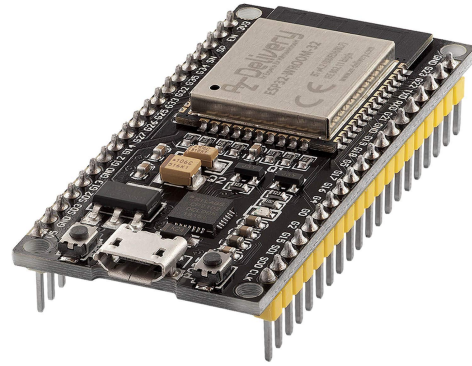


Fig. 2. ESP32

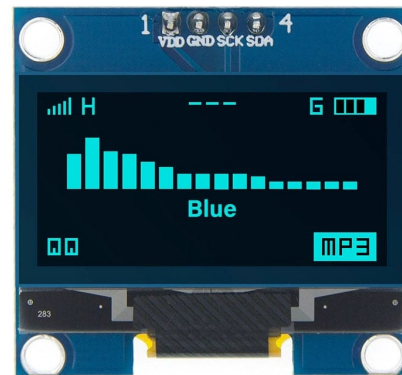


Fig. 3. OLED Display

D. Internet of Things (IoT)

The Internet of Things (IoT) is a network of physical objects that are equipped with sensors, software and technologies that connect and exchange data with other devices and systems over the internet [5]. These devices range from appliances to industrial machinery, collect and share information without direct human intervention. They communicate via Wi-Fi, Bluetooth, cellular or Wide Area Networks (LoRaWAN).



Fig. 4. Internet of Things

E. Ubidots

Ubidots is a cloud-based Internet of Things (IoT) platform that allows businesses and developers to connect devices, collect sensor data, view it on dashboards in real time, and set up alerts. Facilitates the rapid deployment of industrial solutions without the need for complex software development [6]. Ubidots was chosen because of its ease of connecting with the ESP32 and the way the data is displayed.

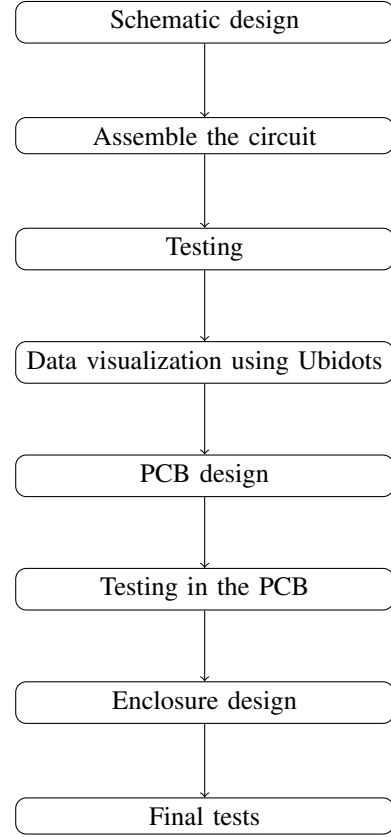


Fig. 5. Ubidots

III. METHODS AND TOOLS

The methodology followed in this project consists of eight structured phases, each aligned with the specific objectives of the project, and ultimately lead to the general objective. The activities are distributed with orderly progression, starting with the basic and then reaching the implementation of the electronic embedded card.

Methodology Flowchart



The tools used to support these activities included EasyEDA (for schematic and PCB design), Arduino (for code development), and Siemens NX (for CAD designs).

IV. DEVELOPMENT

The first step involved testing the sensor with the ESP32. The objective of these tests was to verify the proper operation of the sensor and, at the same time, to define the schematic diagram of the PCB. At the beginning of these tests, only the distance readings between an object and the sensor were considered. To enable communication between the sensor and the ESP32, it is necessary to use a 3.3V to 5V logic level converter, since the ESP32 can only provide 3.3V [1] as an output while the sensor operates at 5V [2]. This converter was used only during the initial tests; in the final version of the prototype, as well as in the PCB design, a voltage divider using two resistors was implemented instead.



Fig. 6. Testing the ultrasonic sensor

Then the measurements were made to the test tank, to determine the behavior of the sensor with diesel. These measurements were successful, which led us to the next step of the project: Schematic design and PCB design. In which the voltage divider with two resistors of 1K and 2K ohm respectively was already used.



Fig. 7. Measurement on the diesel tank

All the data recollected were displayed at the serial monitor in Arduino IDE. The distance were in cm and with this data, it can be implemented to calculate the volume and percentage of the tank diesel.

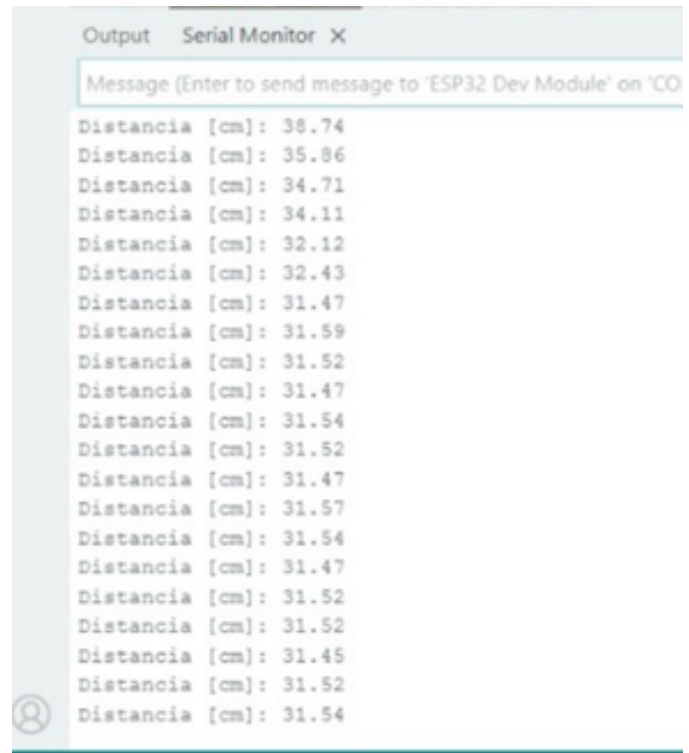
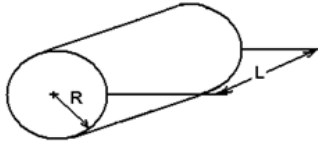


Fig. 8. Measurements shown in the serial monitor

To determine the volume and fill percentage of the tank, it is necessary to know the following parameters: the height, radius, and the distance between the sensor and the tank. As mentioned earlier, the sensor operates within a specific distance range. For this implementation, the reference distance was set to 25 cm, since this is the range at which the sensor performs optimally.

Using this information, the distances corresponding to the full and empty states of the tank, as well as its total capacity, can be established. The tank is considered full when the measured distance is 25 cm. If the measured value is below this threshold, the system will still register it as 25 cm. The tank is considered empty when the measured distance is equal to the full distance plus the total diameter of the tank. Finally, the tank's usable capacity is defined as the difference between the empty and full distance values.

The next thing is to use the formula for a volume of horizontal cylinder:



$$V = L \left[\frac{\pi R^2}{2} + (h - R)\sqrt{2Rh - h^2} + R^2 \arcsin \frac{-R + h}{R} \right]$$

Fig. 9. Volume of a horizontal cylinder

The L correspond to the height of the tank, R for the radius, and h the distance at the level of diesel. For this project the values for these measurements are:

- R = 29.25 cm
- Full = 25 cm
- Empty = 78.5 cm
- L = 87.5 cm
- Capacity = 53.5 cm
- Maximum volume = 225.98 L

Once the volume has been calculated, the next step is to determine the fill percentage. This can be obtained by dividing the calculated volume by the maximum tank capacity and then multiplying the result by 100%.

The calculated percentage value is then displayed on the OLED screen. For this implementation, the libraries `Adafruit_GFX.h` and `Adafruit_SSD1306.h` were used to control the OLED display with the ESP32.

Additionally, another feature was implemented to provide a visual indication of the tank level using an RGB LED. The LED changes color according to the percentage level of the tank. For example, it turns green when the percentage is between 75% and 100%, indicating that the tank is in a good condition. It changes to yellow when the level is between 25% and 75%, representing a stable state. Finally, it turns red when the percentage falls below 25%, indicating a critical condition.

As an additional warning feature, when the tank reaches the critical state, the red LED begins to blink, and a warning image is displayed on the OLED screen for 2 seconds, after which the LED continues blinking to maintain the alert indication. The final test of the final prototype was carried out, in which the OLED screen and RGB LED were implemented in the diesel tank.

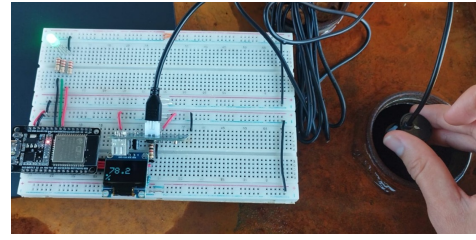


Fig. 10. Test with OLED screen and the LED RGB

To enable remote monitoring of the tank level, the ESP32 was configured to connect to a local Wi-Fi network and transmit the measured data to the Ubidots cloud platform. The ESP32 includes an integrated Wi-Fi module, which allows wireless communication without the need for additional hardware components [7].

The connection process begins by defining the network credentials, specifically the Wi-Fi SSID and password, within the program. Using the `WiFi.h` library, the ESP32 is initialized in station mode and attempts to connect to the selected wireless network. Once the connection is successfully established, the device receives an IP address from the router, allowing internet access and cloud communication.

After connecting to the Wi-Fi network, the ESP32 establishes communication with Ubidots using the MQTT protocol, which is widely used in IoT applications due to its efficiency and low bandwidth consumption. For this implementation, the device authentication token provided by Ubidots was used to securely identify the ESP32 as a registered device within the platform. The measured variables, such as tank volume and percentage level, are then published periodically to Ubidots through the MQTT broker. Once the data is uploaded, Ubidots processes and displays the information in real time through customizable dashboards, allowing continuous monitoring of the tank status from any device with internet access, such as a computer or smartphone. Additionally, the platform can be configured to generate alerts and notifications whenever the tank level reaches critical thresholds.

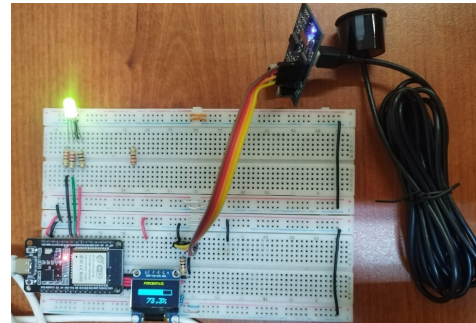


Fig. 11. Final prototype

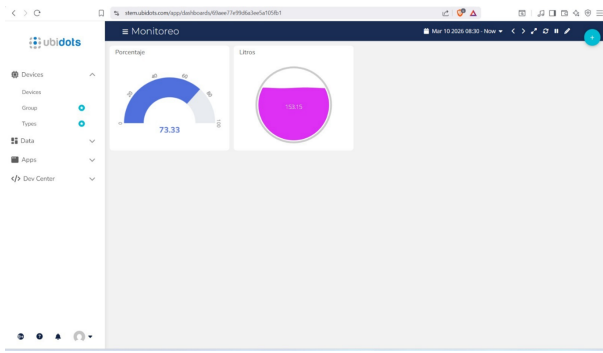


Fig. 12. Data displayed at Ubidots

We now move on to the PCB design section, followed by the testing phase of the PCB with the soldered components. The corresponding tests were carried out on the final PCB using several nearby objects to verify its proper operation and performance.



Fig. 13. Final PCB prototype

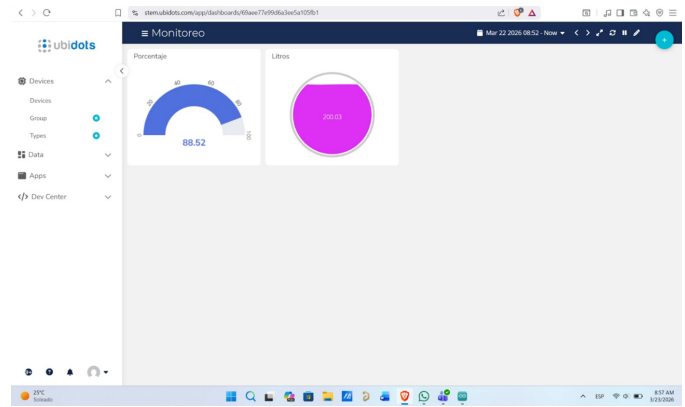


Fig. 14. Testing the final prototype at Ubidots

Finally, the design of the enclosure in which this PCB will be mounted is made. The design was made with the help of the NX tool, in which only the design was made in 3D so that in the future it can be done with a 3D printer and implemented with this PCB.

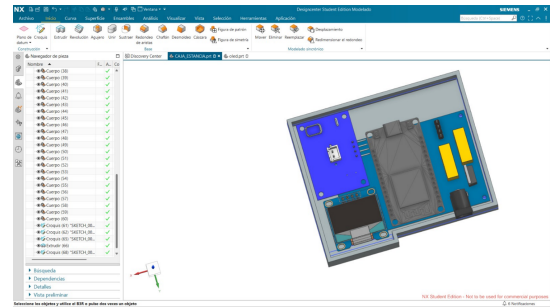


Fig. 15. Enclosure of the PCB

V. RESULTS

The embedded card proved to be a great success, as it meets the requirement: It measures the distance between the diesel level and the sensor. With tank data and measured distance, the card is able to display both the percentage and volume of fuel contained in the tank. In addition, it successfully connects to the internet, which allows this data to be displayed on the Ubidots platform, demonstrating the IoT application offered by this card and its possible uses in the future. Moreover, the card also allows you to program the ESP32 in case you want to modify some parameter, which can be done without any complications, demonstrating the flexibility and versatility of the system.

VI. CONCLUSION

In conclusion, the diesel tank monitoring system represents a highly useful project for both the Centro de Investigación Científica de Yucatán and for us as students, as it allowed us to reinforce and apply the knowledge acquired throughout our academic training.

During the development of the project, it was necessary to pay close attention to every detail in the design, implementation, and testing stages, which helped us understand the importance of planning and timely problem-solving in a real system. Unlike the academic exercises carried out in the classroom, this project involved a more comprehensive process in which each element had to be carefully analyzed and validated to ensure reliable operation.

Furthermore, since the system is intended for an emergency power plant, reliability and monitoring accuracy are critical aspects. Therefore, it was essential to conduct multiple tests and adjustments to guarantee the quality of the final result. In this sense, the project not only met the established objectives but also strengthened our experience in developing engineering solutions for real-world needs.

APPENDIX A

TECHNICAL SPECIFICATIONS OF THE MAIN COMPONENTS

ESP32 Microcontroller

- Operating voltage: 3.3V
- Input voltage (VIN): 5V
- Wireless communication: Wi-Fi and Bluetooth
- ADC resolution: 12 bits
- Supported protocols: UART, I2C, SPI
- Clock frequency: up to 240 MHz

JSN-SR04T Ultrasonic Sensor

- Operating voltage: 5V
- Measuring range: 25 cm to 450 cm
- Frequency: 40 kHz
- Accuracy: ± 3 mm
- Waterproof probe

OLED Display

- Resolution: 128×64 pixels
- Communication protocol: I2C
- Driver: SSD1306
- Operating voltage: 3.3V – 5V

APPENDIX B

PCB CONNECTIONS AND PIN CONFIGURATION

ACKNOWLEDGMENT

I would like to thank CICY for bringing me this opportunity to work in my internship. This experience has enhanced my technical skills and also helped me to realize that working in a enterprise or outside the school, can lead you to a better glance in how to work in the future, the practice that I have seen so far in my major, they are not similar in the practice that I made, and this have motivated me to pursue similar initiatives in the future.

TABLE I
ESP32 PIN CONNECTIONS

Component	ESP32 Pin
Ultrasonic Trigger	GPIO 5
Ultrasonic Echo	GPIO 18
OLED SDA	GPIO 21
OLED SCL	GPIO 22
RGB LED Red	GPIO 25
RGB LED Green	GPIO 26
RGB LED Blue	GPIO 27

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